

Queries Justification

Query 1:

Identifying At-Risk Students (Failing All Subjects)

- Business Goal: Academic Intervention.
- Logic: The school administration needs to identify students who are failing all of their enrolled courses. These students require immediate counseling or tutoring.
- How it works: We search for students who are enrolled in classes but do not have a single grade greater than or equal to 50 (the passing mark). This isolates the most critical cases of academic struggle.

Query 2:

Managing Classroom Resources (Overcrowded Classes)

- Business Goal: Resource Management & Safety Compliance.
- Logic: To ensure safety regulations and effective learning environments, we need to detect classrooms where the number of enrolled students exceeds the physical capacity of the room.
- How it works: The system compares the live count of students assigned to a specific Class_Number against that room's static Capacity limit defined in the Classroom table.

Query 3:

Targeting Multi-Grade Parents

- Business Goal: Targeted Communication.
- Logic: The school often needs to send specific updates to parents who have children in multiple different grade levels (e.g., a "Sibling Discount" offer or a combined scheduling update).
- How it works: We identify parents who are linked to children in more than one unique Academic_Year . We also filter for parents who are "Active" (i.e., they have received announcements in the past), ensuring we don't target inactive profiles.

Query 4:

Recognizing "Star Students" (Dean's List)

- Business Goal: Student Recognition & Awards.
- Logic: To award the "Star Student" status, a student must demonstrate both academic excellence and reliability. High grades alone are not enough; attendance is also mandatory.
- How it works: We calculate the average grade for each student. If the average is 90 or above, we perform a secondary check to ensure the student has never been marked 'Absent' in the Attendance records.